

Candidate-specific paths to dark-matter closure

A strong discovery claim requires I(c), A(c), E(c), and cross-domain identification L(c).

	Constitutive identification	Cosmic attribution	Explanatory adequacy	Cross-domain identification
Laboratory axion/ALP signal	Potentially New particle or field identified	Not by itself Abundance and distribution remain open	Not by itself Cosmic explanatory reach remains open	Not by itself Signal must be linked to cosmic component
WIMP recoil with convergent evidence	Potentially Requires robust signal interpretation	Potentially Requires halo and cosmological inference	Potentially Could account for attributed phenomena	Potentially Convergence could connect all evidence
Cosmologically significant PBH population	Potentially Population identified categorically	Potentially Population and abundance constrained	Potentially Could account for gravitational evidence	Potentially Same population must close the roles
Modified-gravity success	No component No dark constituent identified	No component No dark constituent attributed	Potentially May explain attributed phenomena	No component Resolves or dissolves the composition claim

Assessments indicate what each stipulated result could establish; they are not claims about candidate viability.